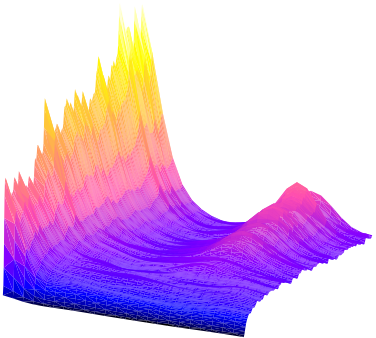
Attention Scaling in Transformer Models



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Introduction

Why do language models get worse as you give them more text?

$$\text{[Input Text]} \xrightarrow{\text{GPT}} \mathbb{P}(\text{Next Token} \mid \text{Input Text})$$

Softmax

The softmax function maps scores to a probability distribution:

$$p_j = \frac{e^{s_j}}{\sum_{k=1}^{n} e^{s_k}}.$$

$$\begin{bmatrix} 1.0 \\ 2.0 \\ 0.5 \\ -1.0 \end{bmatrix} \xrightarrow{\text{softmax}} \begin{bmatrix} 0.24 \\ 0.65 \\ 0.10 \\ 0.01 \end{bmatrix}$$

We can introduce an **inverse temperature** $\beta \geq 0$:

$$p_j(\beta) = \frac{e^{\beta s_j}}{\sum_{k=1}^{n} e^{\beta s_k}}.$$

Low β

More uniform

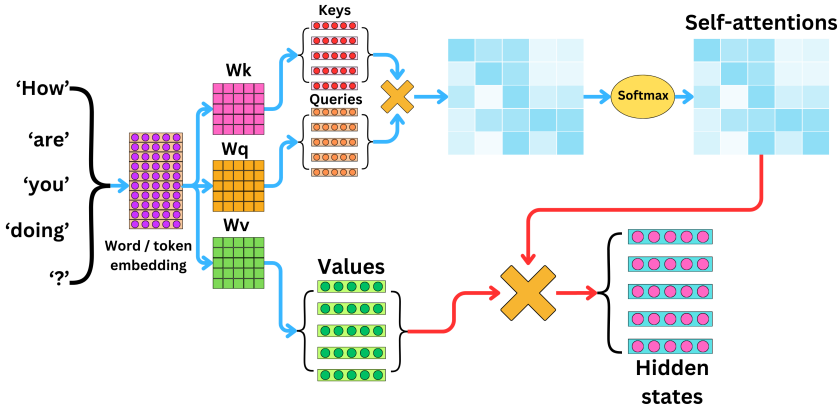
0.27 0.34 0.22 0.17

High β

More concentrated

0.00 0.99 0.01 0.00

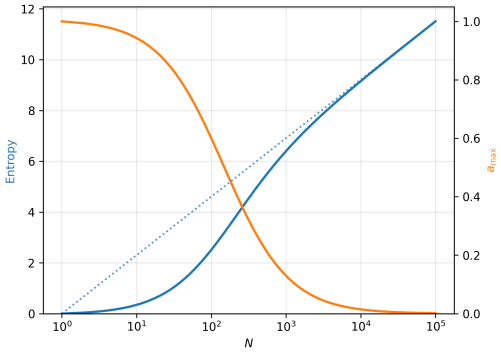
Attention



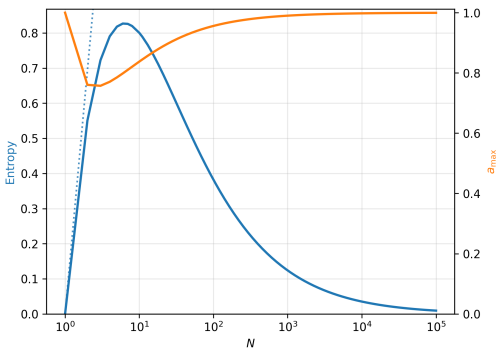
Most models use fixed inverse temperature $\beta = 1$.

Collapse

Under certain conditions,



$\beta_n = O(1)$
rank collapse



$\beta_n \gg$ critical
entropy collapse

Critical Scaling

[Chen et al., 2025]

- Simplex-like token geometry: $\beta_n \sim \log n$

[Giorlandino and Goldt, 2025]

- Randomized weights and inputs: $\beta_n \sim \sqrt{\log n}$

Theorem ([Hayase and Karakida, 2026])

Every $s^{(n)} \in \mathbb{R}^n$ determines a scale $\Lambda_n = \Lambda(s^{(n)}) > 0$ such that

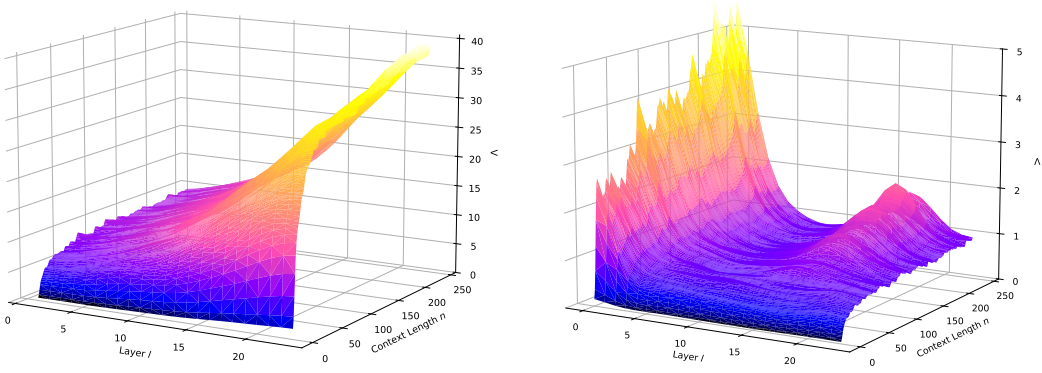
$$\frac{\beta_n}{\Lambda_n} \to \begin{cases} 0, & \textit{rank collapse*} \\ \infty, & \textit{entropy collapse} \end{cases}$$

This quantity is powerful, but under-researched both theoretically and practically.

Personal Work (Theoretical)

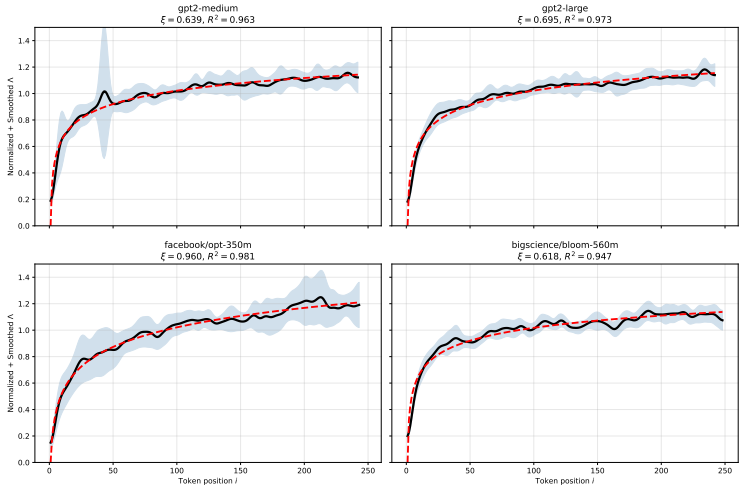
1. Poly-logarithmic critical scaling $\beta_n \sim (\log n)^{\xi}$ isn't universal
2. Any target Λ_n is constructible from a suitable vector
3. Λ_n scales linearly under rescaling ($\Lambda(\beta) = \Lambda/\beta$)
4. Λ_n admits an interpretation as effective scaling

Personal Work (Empirical)



Facebook OPT-350M, before and after training.

Personal Work (Empirical)



Chen, S., Lin, Z., Polyanskiy, Y., and Rigollet, P. (2025).
Critical attention scaling in long-context transformers.

Giorlandino, A. and Goldt, S. (2025).
Two failure modes of deep transformers and how to avoid them: a unified theory of signal propagation at initialisation.

Hayase, T. and Karakida, R. (2026).
A unified framework for critical scaling of inverse temperature in self-attention.